CHEAT SHEET UTS · WEEKS 1–7

MK 6 · Business Analytics

Don't mix these up

- Descriptive (what) → diagnostic (why) → predictive (what next) → prescriptive (what to do).
- First ask: what is one row? The data scale (nominal, ordinal, interval, ratio) decides valid operations.
- Precision = TP/(TP+FP); recall = TP/(TP+FN). Test on unseen data; correlation ≠ causation.
- Optimisation: objective + constraints; linear optimum at a corner; shadow price = value of one more unit of a constraint.
- Marketing analytics: ~385 sample for ±5%; measure promos with a control; chase contribution, not revenue.

W1 Introduction to Business Analytics

What business analytics is · Working principles · Four levels of analytics: harder and more valuable · BetonPrima in charts: four levels in one case

- Four levels.
- DIKW.
- Start from the decision.
- Analytics supports judgement.
- Mislabelled data hides root causes.

W2 How to Deal with Data

One row, one thing · Metadata and data dictionaries · Measurement scales and allowed operations · The shape of a column · Eight defects and who fixes them · One set of sales, four row counts · Column shape: averages can mislead · Three places to fix: cost vs coverage

- What is one row?
- Keys.
- Metadata essentials.
- Scales limit operations.
- Mean needs spread.
- Fix at capture.

W3 Data-Driven Decision Making, Descriptive Analytics & Storytelling

Decisions and DDDM · Descriptive analytics and statistics · Visualisation and storytelling · Intuition and data as partners · Descriptive statistics that belong together · Before and after: from cluttered to a story · For BetonPrima: two charts that convince the board

- Decision types.
- DDDM validates first.
- Data and intuition partner.
- Describe centre, spread, shape.
- Right chart, no clutter.
- Context → finding → action.

FORMULAS

```
Context → finding → action
```

W4 Predictive Analytics

What prediction is, and its kinds · Classification: Titanic · Regression: California housing · What can go wrong · Classification, regression or time series? A quick guide · Titanic: comparing methods and the confusion-matrix calculator · California: how far regression beats the average · Simpson's paradox in a chart

- Three kinds.
- Beat the naive baseline.
- Test on unseen data; CV.
- Precision, recall, F1.
- MAE, RMSE, MAPE, R^2.
- Leakage, drift, Simpson.
- Correlation ≠ causation.

FORMULAS

```
Contoh: 1.730 + 44.476×5,74 + 1.684×17
- 2.815×6,16 = $268.186 (aktual
$218.600, gap $49.586)
```

W5 Prescriptive Analytics: Optimisation

Purpose and anatomy of a decision model · Linear programming: Region 51 · Shadow prices · Scale, integers, diminishing returns · Uncertainty and six traps · Region 51: corner points and simulator · Shadow prices in a chart · Scale, integers and diminishing returns · Platelet stock: asymmetric costs

- Data shows the pain; decisions need objective and constraints.
- LP optimum at a corner.
- Shadow prices.
- Integer constraints, don't round.
- The benefit curve shapes the plan.
- Weigh the two errors.

FORMULAS

```
max 6D + 1,2B   s.t. 0,8D + 0,12B ≤ 40; D + 0,5B ≤ 78; D ≤ 45; B ≤ 129; D, B ≥ 0
```

```
Critical ratio = biaya kurang ÷ (biaya kurang + biaya lebih) = 2,5 ÷ 3,1 = 0,81
```

```
CR = 2.5/3.1 = 0.81 → ~122 units
```

W6 Marketing Analytics

Framing and measurement foundations · Customer analytics · Promotion and pricing · Marketing Mix Modeling · Location analytics and analysis to decision · Sample size: why ~385 works for a city or a country · RFM: from scores to actions · Optimal price, elasticity and market basket · MMM in charts: adstock and reallocation · Eiger vs Torch and the coffee war

- Decision, data, confidence.
- Design before sample size.
- RFM, CLV, survival.
- Causal impact needs a control.
- Optimise contribution, not revenue.
- MMM with adstock.
- AI is not a source of numbers.

FORMULAS

```
n₀ = z² · p(1-p) / MoE²; n = n₀ / (1 + (n₀ - 1)/N)
```

```
CLV = Σ m · rᵗ / (1 + d)ᵗ
```

```
n₀ ≈ 385 for ±5% at 95%
```

W7 Financial Analytics

Monthly sales forecast for the 2027 budget · Nowcasting the current month · Credit scoring · The 2027 budget forecast in charts · Three ways to estimate October on the 12th · Credit scoring: methods and cut-offs in charts

- Finance uses of analytics.
- Lebaran columns cut error to 3.2%.
- Moving holidays need their own columns.
- Calendar nowcasts.
- Models rank; managers set cut-offs.
- Sensitive features are policy choices.